



Lesson 3.6 — Polynomial Division

Big idea: When you divide one polynomial by another, you get a quotient and a remainder — just like integer division. Long division always works; synthetic division is a fast shortcut for divisors of the form $(x - c)$.

Quick review

Division algorithm: $P(x) = D(x) \cdot Q(x) + R(x)$, where $\deg(R) < \deg(D)$.

Long division: works for any divisor. Subtract step-by-step until the remainder is lower degree than the divisor.

Synthetic division: shortcut when dividing by $(x - c)$. Bring down, multiply by c , add, repeat. Last number = remainder.

Use a place-holder zero for any missing power in P .

Worked example

Worked example. Divide $(x^3 - 2x^2 - 5x + 6)$ by $(x - 1)$ using synthetic division.

$$\begin{array}{r|rrrr} 1 & 1 & -2 & -5 & 6 \\ & & 1 & -1 & -6 \\ \hline & 1 & -1 & -6 & 0 \end{array}$$

Quotient: $x^2 - x - 6$. Remainder: 0. (So $x - 1$ is a factor.)

Warm-ups

1. Set up: divide $x^3 - 8$ by $(x - 2)$. What is the synthetic divisor c ?
2. Divide $(x^2 + 5x + 6)$ by $(x + 2)$.
3. If $P(x)$ divided by $(x - 3)$ leaves remainder 7, what is $P(3)$?

Practice

1. Divide $(2x^3 + 3x^2 - x - 6)$ by $(x + 2)$ synthetically.
2. Divide $(x^4 - 16)$ by $(x - 2)$.
3. Use long division: divide $(x^3 + 2x^2 - 5x - 6)$ by $(x^2 - 1)$.

Challenge

1. Divide $(3x^4 - 5x^2 + 4)$ by $(x - 1)$. State quotient and remainder.
2. Factor $P(x) = x^3 - 7x + 6$ completely, given that $x = 1$ is a zero.



Answer key — Lesson 3.6

Check your work. If a problem tripped you up, rewatch the step in the video where that concept appears.

Warm-ups

1. $c = 2$.
2. Quotient: $x + 3$. Remainder: 0.
3. $P(3) = 7$ (Remainder Theorem).

Practice

1. $-2 \mid 2 \quad 3 \quad -1 \quad -6 \rightarrow$ bring down 2; $2(-2) = -4$, $3 + (-4) = -1$; $-1(-2) = 2$, $-1 + 2 = 1$; $1(-2) = -2$, $-6 + (-2) = -8$. Quotient: $2x^2 - x + 1$, remainder -8 .
2. Synthetic with $c = 2$ on coefficients 1 0 0 0 -16 : row 2, 4, 8, 16; sum 1 2 4 8 0. Quotient: $x^3 + 2x^2 + 4x + 8$, remainder 0.
3. $x^2 - 1$ goes into $x^3 + 2x^2 - 5x - 6$: x times $\rightarrow x^3 - x$; subtract $\rightarrow 2x^2 - 4x - 6$; $x^2 - 1$ goes 2 times $\rightarrow 2x^2 - 2$; subtract $\rightarrow -4x - 4$. Quotient: $x + 2$, remainder $-4x - 4$.

Challenge

1. Coefficients 3 0 -5 0 4. $c = 1$: row 3, 3, -2 , -2 ; sums 3 3 -2 -2 2. Quotient: $3x^3 + 3x^2 - 2x - 2$, remainder 2.
2. Synthetic on 1 0 -7 6 with $c = 1$: row 1, 1, -6 ; sums 1 1 -6 0. Quotient $x^2 + x - 6 = (x + 3)(x - 2)$. So $P(x) = (x - 1)(x + 3)(x - 2)$.